The AI plans.
You authorize.
The contract enforces.

AgentWallet: a policy-bounded AI treasury agent, executed on Ethereum Sepolia.

PUBLIC TESTNET · ETHEREUM SEPOLIA

0x8A44716Efabb6Ce401291B91D4427149CaF956a6

11 / 11 transactions SUCCESS

THE PROBLEM

Give an AI a wallet, and it can lose the money.

Hallucination

The model invents a recipient or an amount, and nobody notices before it's signed.

Prompt injection

Text hidden in a label, memo or web page redirects the payment.

Runaway autonomy

A scheduled agent keeps spending after conditions change, with no owner in the loop.

Asking the model to be careful is not a control. The controls have to sit **outside** the model.

Every payment crosses four gates.

1 · AI plans

gpt-4.1-mini turns plain English into typed tool calls. It has no tool that can sign, approve or loosen.

2 · Policy checks

14 block-pinned checks, including a pre-flight simulation, in trusted code on every path (15 for task runs).

3 · Owner authorizes

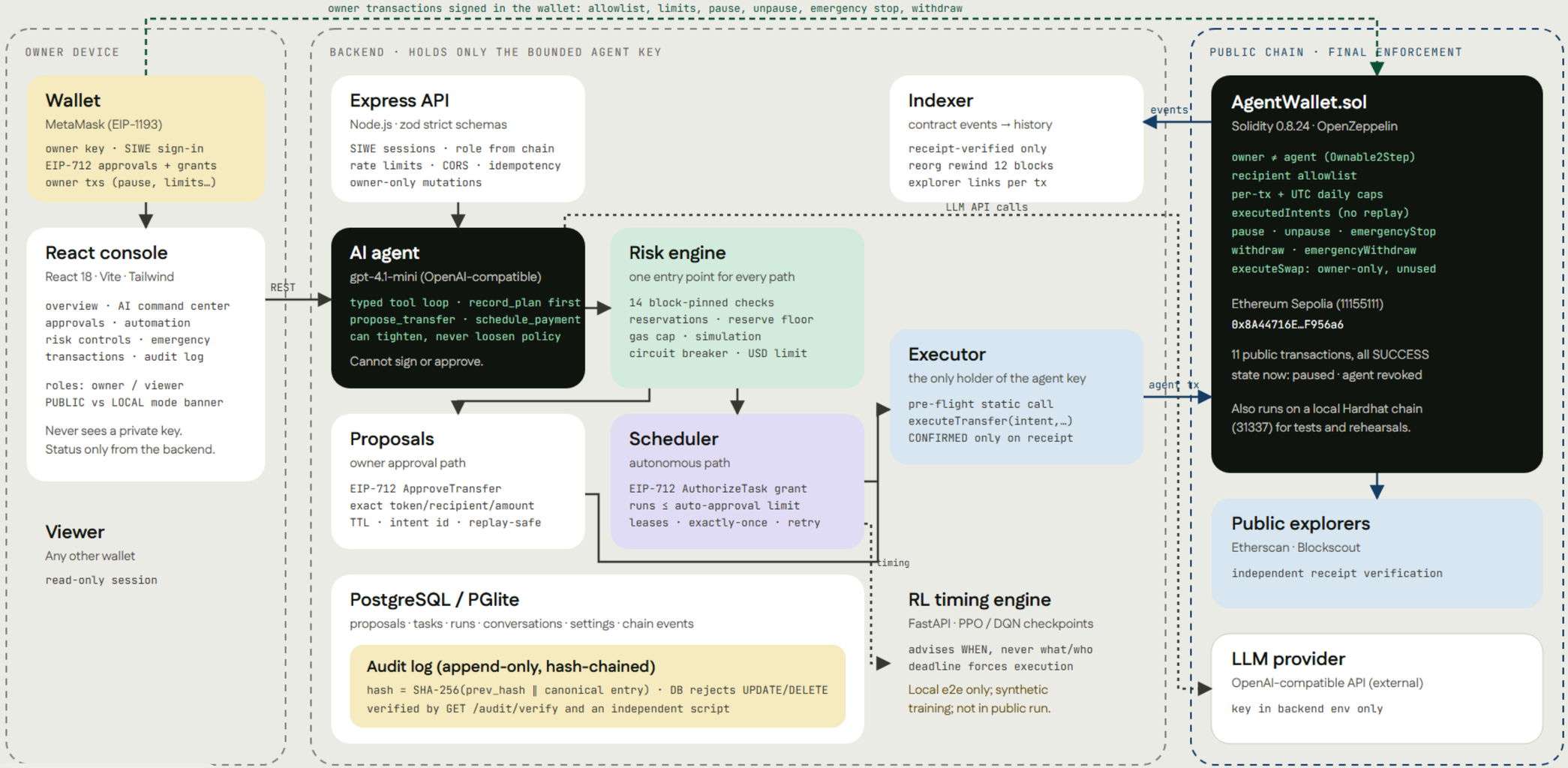
EIP-712 signature per transfer, or one bounded task grant with an auto-approval limit.

4 · Contract enforces

Allowlist, per-tx and daily caps, one-time intents, pause and emergency stop, on-chain.

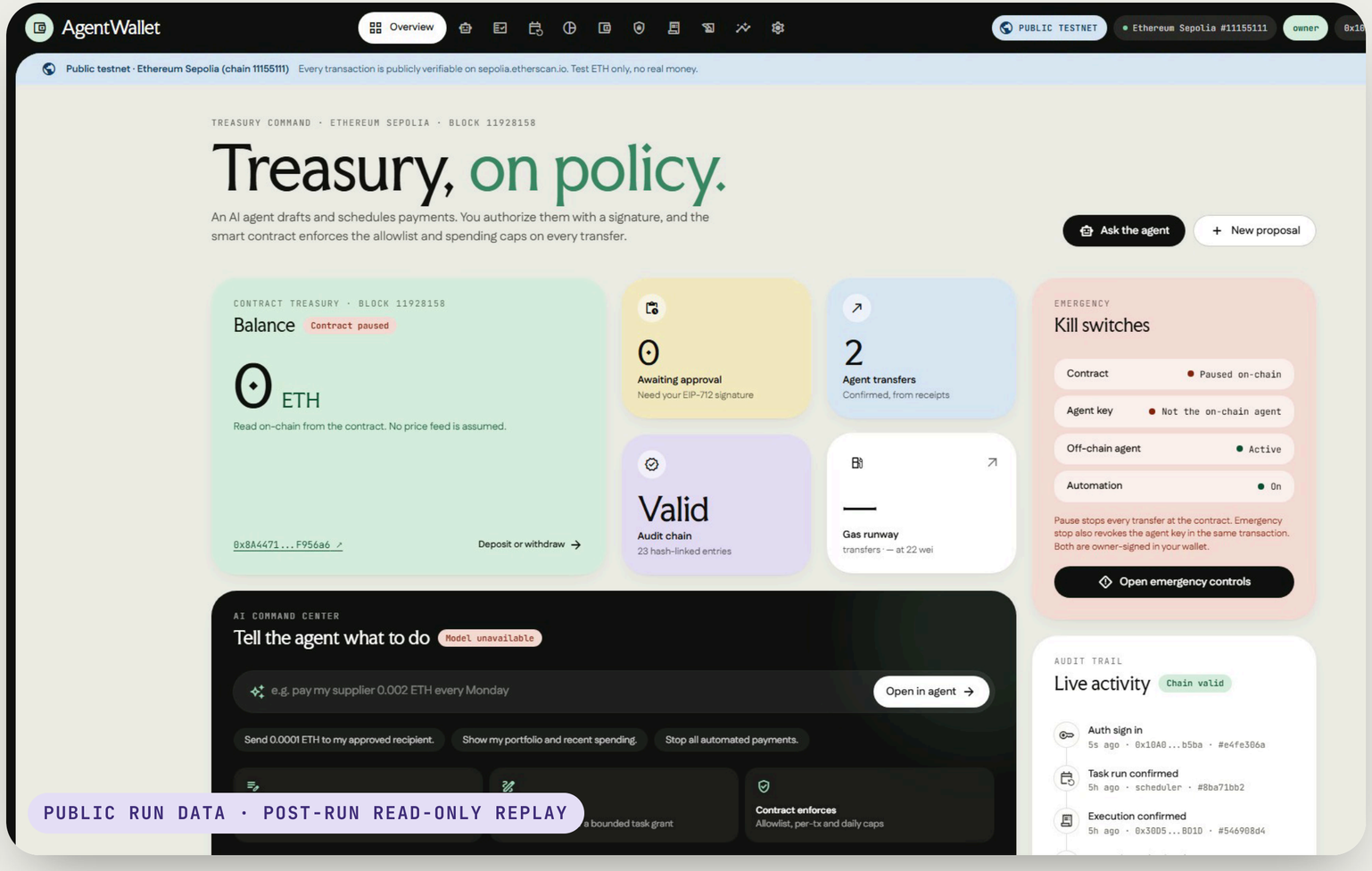
AgentWallet: system architecture

The AI plans · the owner authorizes · the backend risk engine checks · the smart contract enforces. Dashed boxes are trust boundaries.



ARCHITECTURE DIAGRAM

A treasury command deck.



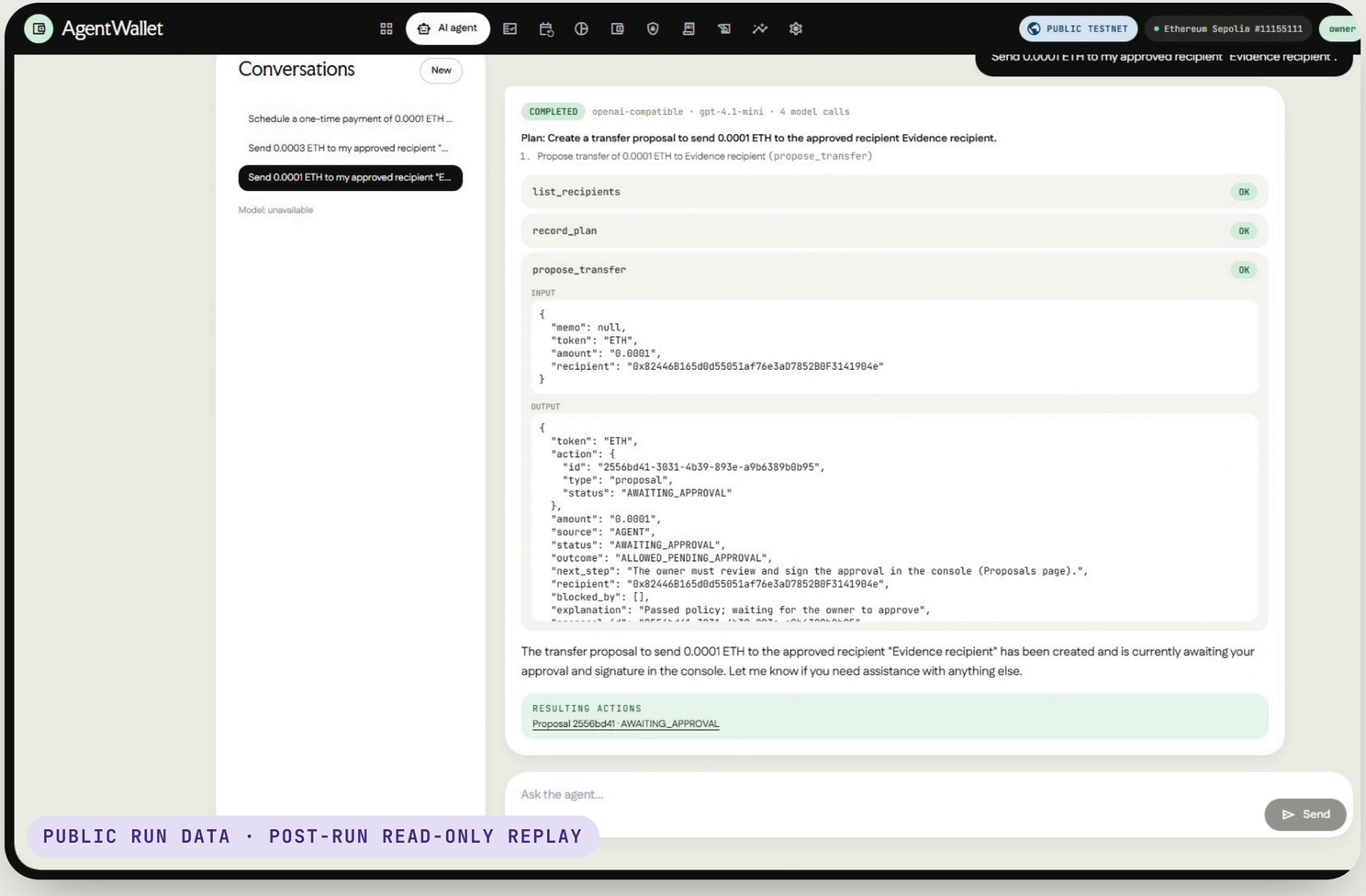
Public vs local, always visible

A sky PUBLIC TESTNET banner, or a hatched LOCAL SIMULATION banner. Explorer links only for public transactions.

Everything from the chain

Balance, caps, approvals, agent transfers, audit validity and kill switches, read from the backend and the contract.

Plain English in, a draft out.



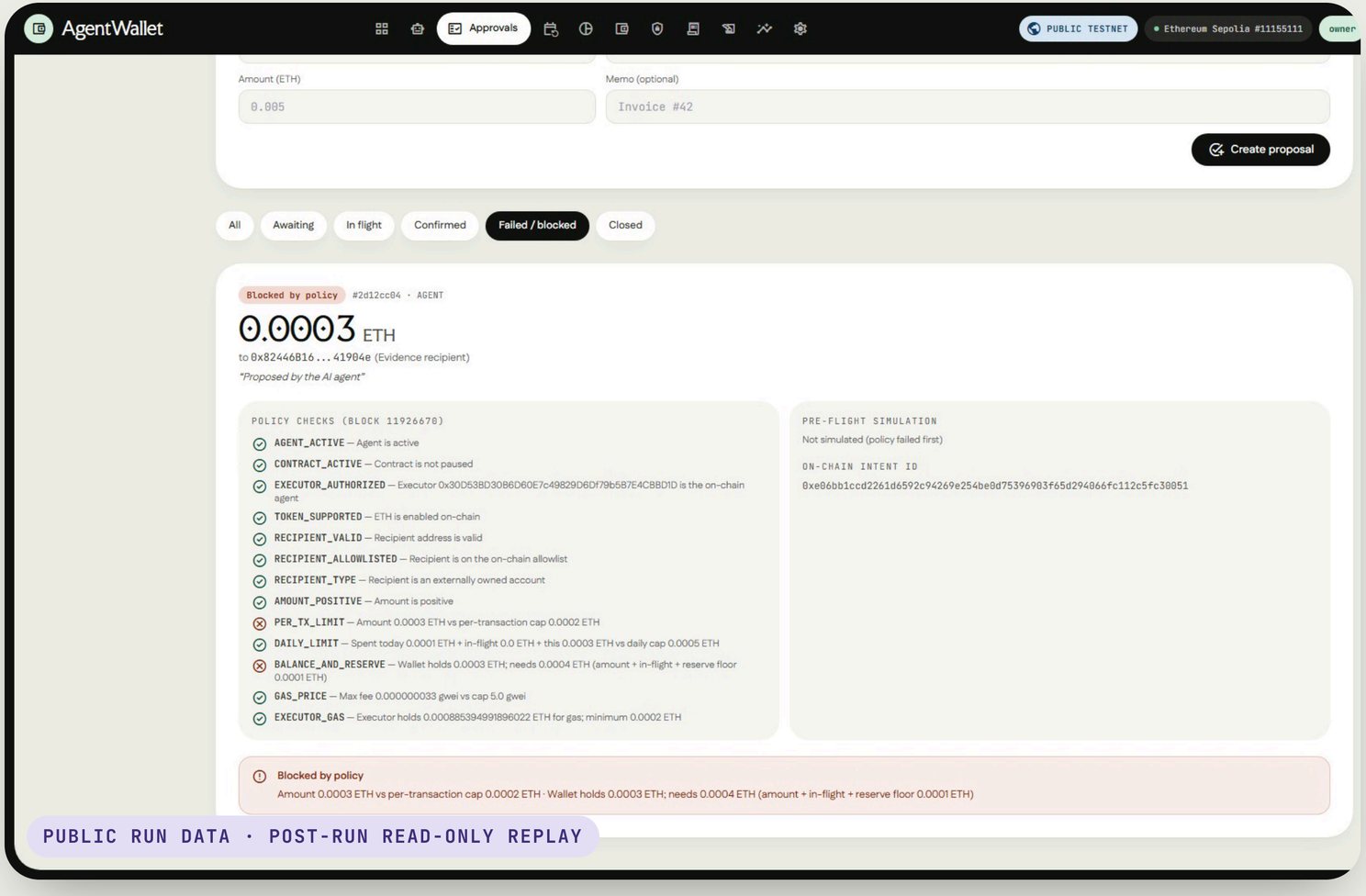
Real turn, public run

`list_recipients → record_plan → propose_transfer → AWAITING_APPROVAL`

Guardrails in code

- recipients only from the on-chain allowlist
- action tools only after `record_plan`
- tighten-only policy tools
- labels and memos treated as data

The AI drafted 0.0003 ETH. Policy said no.



Failed checks

```
PER_TX_LIMIT: 0.0003 > 0.0002
BALANCE_AND_RESERVE: needs 0.0004,
holds 0.0003
```

Nothing sent

Never signed, never broadcast. Audit entry #10 `PROPOSAL_POLICY_REJECTED` has no transaction hash. The agent key sent exactly 2 transactions in the whole run.

AUTHORIZATION

One signature per transfer, or one grant for many.

Owner approval · EIP-712 ApproveTransfer

Signs the exact proposal ID, intent ID, token, recipient, amount and deadline. Policy is re-checked at approval and again before broadcast.

`public: 0x2c83a855…2e37fb · block 11926668`

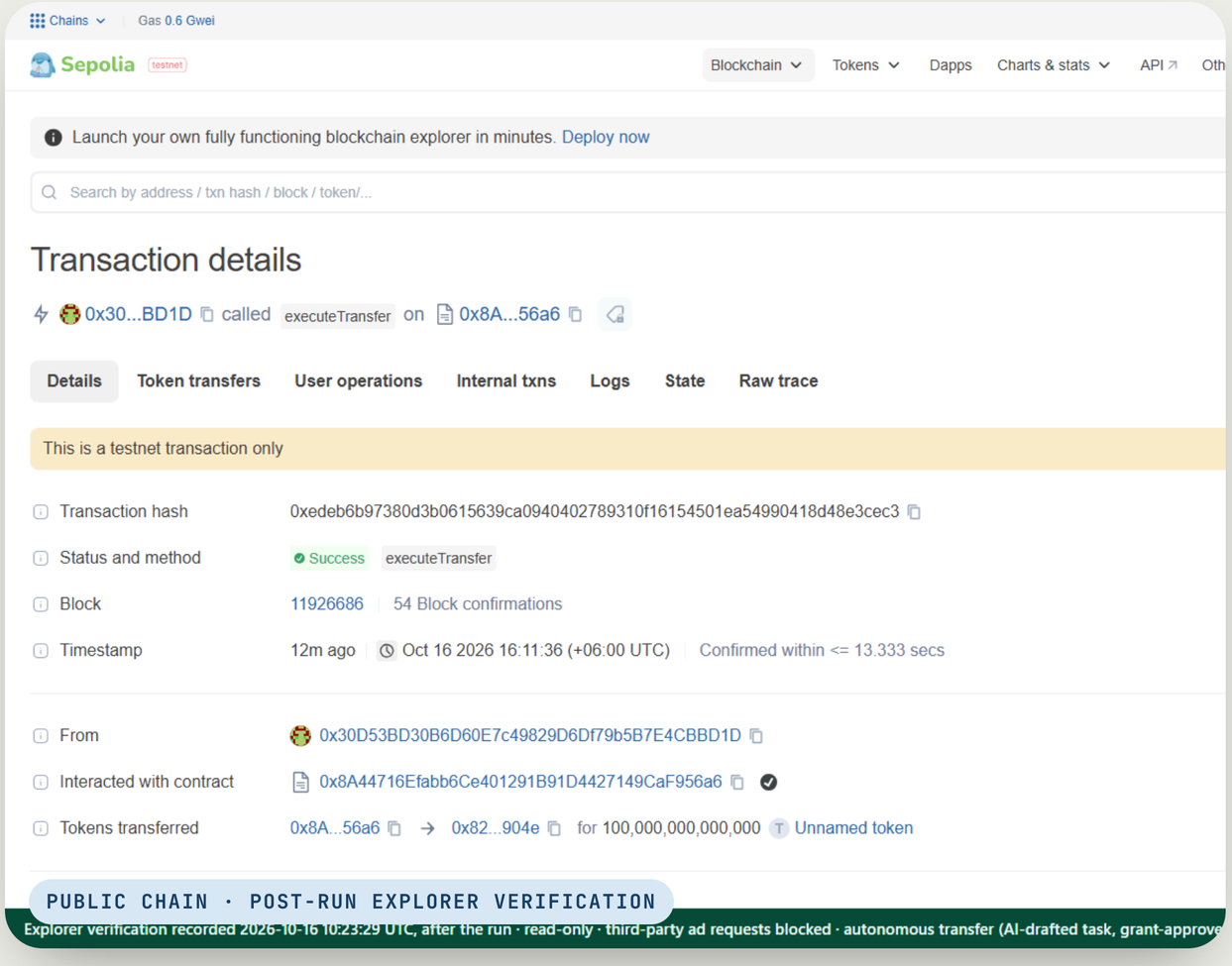
Bounded grant · EIP-712 AuthorizeTask

Fixes recipient, amount per run, max runs, max total, validity and schedule. Runs execute without a signature **only** under the owner's auto-approval limit; otherwise they escalate.

`public: 0xedeb6b97…e3cec3 · block 11926686`

Public signing used a test-only EIP-1193 wallet bridge. Real MetaMask 13.52.0 signing was verified on a local chain (clip 12).

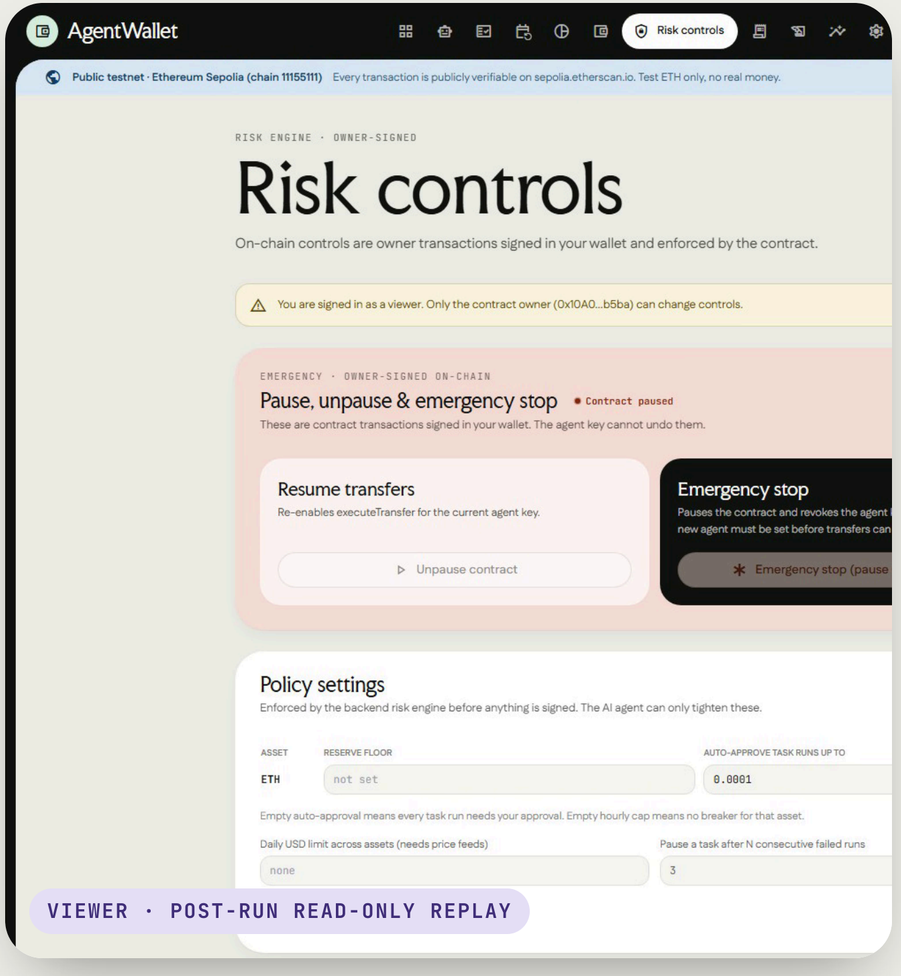
The agent paid on its own, inside a grant.



UTC	STEP
10:08:55	gpt-4.1-mini drafts the task (`schedule_payment`)
10:09:07	**Owner signs the grant, once**
10:11:21	Scheduler creates the run
10:11:26	Policy passes · auto-approved by the grant
10:11:29	Agent key submits `executeTransfer`
10:11:36	**Mined: block 11926686, Success**

Who can do what, and who enforces it.

TIER	CAN	ENFORCED
T0 Anonymous	config, nonce	backend
T1 Viewer	read only (403 on writes)	backend
T2 LLM agent	read, draft, tighten	backend
T3 Scheduler	runs inside a signed grant	backend + contract
T4 Agent key	`executeTransfer` only	contract
T5 Owner	limits, allowlist, pause, stop, withdraw	contract



OWNER OVERRIDE · ON-CHAIN

Stop everything in one transaction.



Pause

`0x2a2eeb50…bd211b`
`block 11926689`



Unpause

`0x00b0dc31…f2f3d0`
`block 11926691`



Emergency stop

`0xfe627133…e53a8d`
`Paused + AgentUpdated + EmergencyStop`



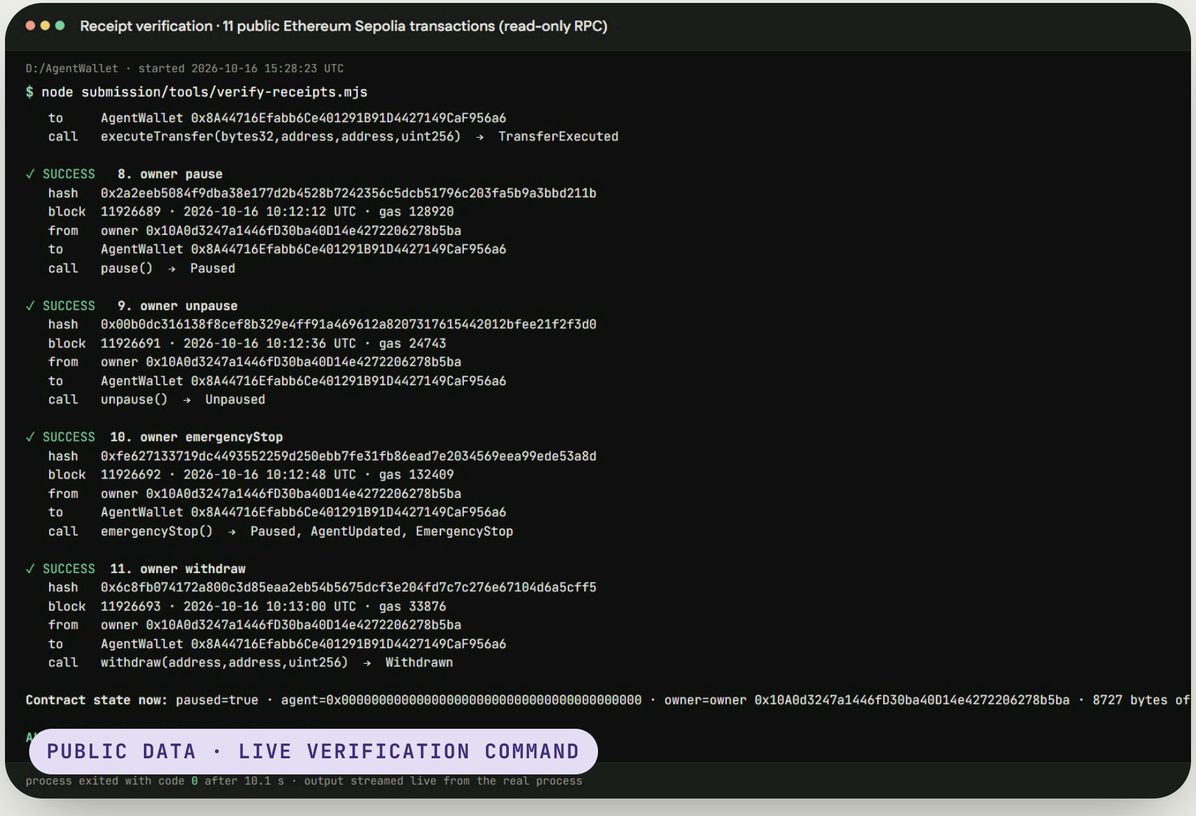
Withdraw

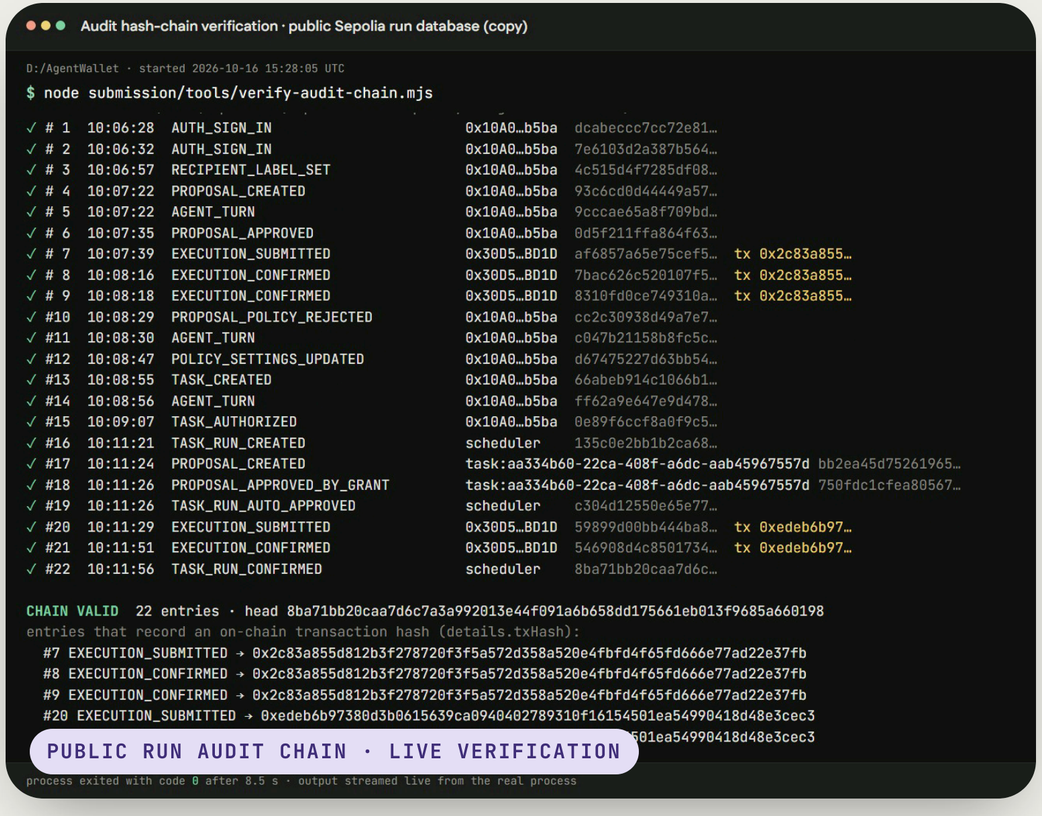
`0x6c8fb074…a5cff5`
`block 11926693`

State today, read-only: `paused = true`, `agent = 0x0`. The contract is still stopped.

EXECUTION EVIDENCE

11 public transactions. All successful.





Independent receipt check over a public RPC, and an independent recomputation of the 22-entry audit chain, which correlates AI decisions with transaction hashes.

VERIFICATION

Tested like it holds money.

35
contract tests: re-entrancy, cap fuzzing, owner/agent separation

118
backend tests: auth, races, scheduler, agent safety

24+3
UI component tests + browser flows

8
full-stack end-to-end on a real local chain

10/11
real MetaMask 13.52.0 rows, local chain (row 9 needs a paid LLM call)

19×4
live gpt-4.1-mini safety assertions

11/11
public transactions re-verified

0
secrets found in committable files

What we don’t claim.

Limits

- Public signing used a test wallet bridge; MetaMask is local
- ETH only on Sepolia; ERC-20 and USD valuation are local (synthetic feeds)
- RL timing is local and synthetic-trained
- Audit chain is off-chain (hash-chained, not anchored)
- Self-assessment only: no external audit

Next

- **Policy-bounded rebalancing:** AI proposes, owner signs, using the `executeSwap` already deployed (not built yet)
- Anchor the audit head on-chain
- Multisig or hardware-wallet owner
- Recurring autonomy on a public chain

Autonomous where it's safe. Human where it matters.

Contract (Ethereum Sepolia)

0x8A44716Efabb6Ce401291B91D4427149CaF956a6

Autonomous transaction

0xedeb6b97…e3cec3

Code

github.com/Turjo101365/AgentWallet